

Indices and surds - Set 2 - Solutions

Specification MM1.1, MM1.2, M2.7, M2.11, M2.12.

Answer key

Question	Answer	Topic
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Worked solutions

Question 1 IND-20 · Indices and surds · spec MM1.2

Given that $\sqrt{a} - \sqrt{b} = 2$ and $a - b = 12$, find $\sqrt{a} + \sqrt{b}$.

Answer: D

$$a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b}), \text{ so } 12 = 2(\sqrt{a} + \sqrt{b}), \text{ giving } 6.$$

Full method - how and why

You are given a difference of surds and a difference of the numbers. The link is the difference of two squares - no need to find a and b .

- Factorise $a - b$ as a difference of two squares in $\sqrt{a}$ and $\sqrt{b}$: $a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$.
why: $a = (\sqrt{a})^2$ for $a \geq 0$, so the whole expression is $X^2 - Y^2$ with $X = \sqrt{a}$, $Y = \sqrt{b}$ - this is the only structural idea in the question.
- Substitute the given values: $12 = 2 \times (\sqrt{a} + \sqrt{b})$.
why: The first bracket is exactly the quantity given as 2.
- Divide: $\sqrt{a} + \sqrt{b} = 6$.
why: Direct.

Where it goes wrong

- Solving simultaneously for a and b (possible: $a = 16$, $b = 4$) - correct but three times the work and easy to fumble.
- Answering 24 by multiplying rather than dividing.

Takeaway $a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$ converts between sums and differences of surds instantly.

Question 2 IND-21 · Indices and surds · spec MM1.1

If $x = 2^p$, $y = 2^q$ and $\frac{x^3}{y^2} = 2^5$ with $xy = 2^{10}$, find p .

Answer: F

$$3p - 2q = 5 \text{ and } p + q = 10. \text{ From the second, } q = 10 - p; \text{ then } 3p - 2(10 - p) = 5, \text{ so } 5p = 25 \text{ and } p = 5.$$

Full method - how and why

Two equations in disguise. Writing both conditions in base 2 turns them into simultaneous linear equations in p and q .

- $\frac{x^3}{y^2} = \frac{2^{3p}}{2^{2q}} = 2^{3p-2q}$, so $3p - 2q = 5$.
why: Powers of powers multiply indices, division subtracts them; equating indices is valid because base 2 is one-to-one.
- $xy = 2^p \cdot 2^q = 2^{p+q}$, so $p + q = 10$.
why: Multiplication adds indices.
- Solve: from the second, $q = 10 - p$; substituting, $3p - 2(10 - p) = 5 \Rightarrow 5p = 25 \Rightarrow p = 5$.
why: Substitution is quickest with only two unknowns; note the bracket when substituting a negative.

Where it goes wrong

- Dropping the bracket to get $3p - 20 - 2p = 5$ and hence $p = 25$.
- Equating $x^3/y^2 = 2^5$ to $3p/2q = 5$ - indices subtract, they do not divide.

Takeaway Converting everything to one base turns exponential conditions into linear equations in the exponents.

Question 3 IND-06 · Indices and surds · spec MM1.2

Given that $(2 + \sqrt{3})^2 = a + b\sqrt{3}$ and $(2 + \sqrt{3})^{-1} = c + d\sqrt{3}$, where a, b, c, d are integers, find the value of $a + b + c + d$.

Answer: E

$$(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}, \text{ so } a = 7, b = 4. (2 + \sqrt{3})^{-1} = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3}, \text{ so } c = 2, d = -1. \text{ Sum} \\ = 7 + 4 + 2 - 1 = 12.$$

Full method - how and why

Two standard surd manipulations in one: squaring a bracket and rationalising a reciprocal. The question then just asks you to keep track of the four integers.

- 1 Square the bracket: $(2 + \sqrt{3})^2 = 4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$, so $a = 7, b = 4$.
why: Use $(x + y)^2 = x^2 + 2xy + y^2$ with $y = \sqrt{3}$, and remember $(\sqrt{3})^2 = 3$ - that is what makes the rational part $4 + 3$.
- 2 Rationalise the reciprocal: $\frac{1}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3}$, so $c = 2, d = -1$.
why: The conjugate gives denominator $2^2 - 3 = 1$, which is why this particular reciprocal is so clean.
- 3 Add: $7 + 4 + 2 + (-1) = 12$.
why: The question wants the sum, so the sign of d matters.

Where it goes wrong

- Taking $d = 1$ and getting 14 - the rationalised form is $2 - \sqrt{3}$, so d is negative.
- Writing $(2 + \sqrt{3})^2 = 4 + 3 = 7$ (forgetting the cross term $2 \times 2 \times \sqrt{3}$).

Takeaway $(a + \sqrt{b})$ and $(a - \sqrt{b})$ multiply to the integer $a^2 - b$; that single fact drives both halves of this question.

Question 4 IND-07 · Indices and surds · spec MM1.1

Which of the following is equal to $\frac{x^2\sqrt{x}}{\sqrt[3]{x}}$ for $x > 0$?

Answer: F

$$\text{The index is } 2 + \frac{1}{2} - \frac{1}{3} = \frac{12 + 3 - 2}{6} = \frac{13}{6}.$$

Full method - how and why

Roots disguised as indices. Convert every root to a fractional power and the question becomes adding and subtracting fractions.

- 1 Rewrite the roots: $\sqrt{x} = x^{1/2}$ and $\sqrt[3]{x} = x^{1/3}$.
why: $\sqrt[n]{x} = x^{1/n}$. In index form the laws apply; in root form they do not.
- 2 Combine: $\frac{x^2 \cdot x^{1/2}}{x^{1/3}} = x^{2 + \frac{1}{2} - \frac{1}{3}}$.
why: Multiplication adds indices, division subtracts them - the three operations become one sum.
- 3 Add the fractions over a common denominator 6: $2 + \frac{1}{2} - \frac{1}{3} = \frac{12 + 3 - 2}{6} = \frac{13}{6}$.
why: 6 is the lowest common denominator of 1, 2 and 3; converting 2 to $\frac{12}{6}$ is where slips happen.

Where it goes wrong

- Adding the $\frac{1}{3}$ instead of subtracting gives $x^{17/6}$, which is offered.
- Forgetting to convert the whole number 2 into sixths.

Takeaway Turn every root into a fractional index immediately; then all you are doing is arithmetic with fractions.

Question 5 IND-09 · Indices and surds · spec MM1.2

Find the value of $\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{2+\sqrt{3}}$.

Answer: D

Rationalising each term gives $(\sqrt{2}-1) + (\sqrt{3}-\sqrt{2}) + (2-\sqrt{3})$, which telescopes to 1.

Full method - how and why

A telescoping sum. Each term rationalises to a difference of two surds, and almost everything cancels - so do not compute decimals.

1 Rationalise each term: $\frac{1}{\sqrt{2}+1} = \sqrt{2}-1$, $\frac{1}{\sqrt{3}+\sqrt{2}} = \sqrt{3}-\sqrt{2}$, $\frac{1}{2+\sqrt{3}} = 2-\sqrt{3}$.

why: Each denominator has the form $\sqrt{n+1} + \sqrt{n}$ whose conjugate gives $(\sqrt{n+1} + \sqrt{n})(\sqrt{n+1} - \sqrt{n}) = 1$ - the denominators all become 1, which is the point of the construction.

2 Add them: $(\sqrt{2}-1) + (\sqrt{3}-\sqrt{2}) + (2-\sqrt{3})$.

why: Write the terms in a line so the cancelling pairs are visible.

3 Cancel: $\sqrt{2}$ with $-\sqrt{2}$, $\sqrt{3}$ with $-\sqrt{3}$, leaving $-1 + 2 = 1$.

why: This is a telescoping sum - the surds annihilate in pairs and only the two ends survive.

Where it goes wrong

- Approximating numerically and choosing $\sqrt{2}$ or $\sqrt{3}$ because the total 'looks about right'.
- Rationalising with the wrong conjugate sign, which flips one term and destroys the cancellation.

Takeaway $\frac{1}{\sqrt{n+1} + \sqrt{n}} = \sqrt{n+1} - \sqrt{n}$: recognise this and long chains of such fractions collapse instantly.

Question 6 IND-11 · Indices and surds · spec MM1.2

Given that $x = \sqrt{7} + \sqrt{3}$ and $y = \sqrt{7} - \sqrt{3}$, find $x^2 + y^2 - xy$.

Answer: B

$x^2 + y^2 = (10 + 2\sqrt{21}) + (10 - 2\sqrt{21}) = 20$ and $xy = 7 - 3 = 4$. So $20 - 4 = 16$.

Full method - how and why

A symmetric surd expression. Computing x^2 , y^2 and xy separately is much cleaner than expanding everything, because the irrational parts cancel.

1 $x^2 = (\sqrt{7} + \sqrt{3})^2 = 7 + 2\sqrt{21} + 3 = 10 + 2\sqrt{21}$ and $y^2 = 10 - 2\sqrt{21}$.

why: The cross terms have opposite signs because y is the conjugate of x - that is what makes the sum rational.

2 Add: $x^2 + y^2 = 20$.

why: The $\pm 2\sqrt{21}$ terms cancel exactly.

3 $xy = (\sqrt{7} + \sqrt{3})(\sqrt{7} - \sqrt{3}) = 7 - 3 = 4$.

why: Difference of two squares again - no cross terms at all.

4 Combine: $x^2 + y^2 - xy = 20 - 4 = 16$.

why: Subtract in the order the question asks.

Where it goes wrong

- Computing $x^2 + y^2$ as $(x + y)^2$ and getting 28 - remember $(x + y)^2 = x^2 + y^2 + 2xy$.
- Giving $20 - 2\sqrt{21}$ by forgetting that the surds cancel when the squares are **added**.

Takeaway With conjugate pairs, $x + y$, xy and $x^2 + y^2$ are all rational; build the expression from those rather than expanding blindly.

Question 7 IND-16 · Indices and surds · spec MM1.2

Which of the following statements is/are true for all positive real numbers a and b ?

- I $\sqrt{a}\sqrt{b} = \sqrt{ab}$
II $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$
III $\sqrt{a^2 + b^2} < a + b$

Answer: F

I is a law of surds. II fails for $a = b = 1$: $\sqrt{2} \neq 2$. III: $(a + b)^2 = a^2 + b^2 + 2ab > a^2 + b^2$ since $ab > 0$, and both sides are positive, so III is true.

Full method - how and why

Three surd 'rules', two of which are genuine. The exam skill is knowing which laws distribute and finding a counterexample for the one that does not.

- 1 Statement I: $\sqrt{a}\sqrt{b} = \sqrt{ab}$ is a law of surds, valid for all $a, b \geq 0$. **True.**

why: Roots distribute over multiplication and division, and the condition $a, b > 0$ in the question is exactly what the law needs.

- 2 Statement II: test $a = b = 1$: $\sqrt{1+1} = \sqrt{2} \approx 1.41$ but $\sqrt{1} + \sqrt{1} = 2$. **False.**

why: One counterexample is enough to kill a 'for all' statement - and the simplest numbers usually work.

- 3 Statement III: square both sides. $(a + b)^2 = a^2 + b^2 + 2ab > a^2 + b^2$ since $ab > 0$, and both sides are positive, so $\sqrt{a^2 + b^2} < a + b$. **True.**

why: Squaring is safe here precisely because both sides are known to be positive; that is what makes the comparison valid in both directions.

- 4 I and III are true: the answer is 'I and III only'.

why: Match your three verdicts to the option list rather than reasoning about the options themselves.

Where it goes wrong

- Assuming roots distribute over addition because they distribute over multiplication - II is the classic error this question is built around.
- Rejecting III because the triangle-inequality picture feels like it should be an equality.

Takeaway Roots split over $\times$ and $\div$ but never over $+$ or $-$; to compare positive quantities, square them.

Question 8 IND-18 · Indices and surds · spec MM1.2

The expression $\frac{3}{7 - 2\sqrt{3}} - \frac{3}{7 + 2\sqrt{3}}$ simplifies to

Answer: F

Common denominator $(7 - 2\sqrt{3})(7 + 2\sqrt{3}) = 49 - 12 = 37$. Numerator: $3(7 + 2\sqrt{3}) - 3(7 - 2\sqrt{3}) = 12\sqrt{3}$.

Full method - how and why

A difference of two rationalisable fractions. Using a single common denominator is faster than rationalising each one.

- 1 Use the common denominator $(7 - 2\sqrt{3})(7 + 2\sqrt{3}) = 49 - 12 = 37$.

why: The two denominators are conjugates, so their product is automatically rational - no separate rationalising needed.

- 2 Numerator: $3(7 + 2\sqrt{3}) - 3(7 - 2\sqrt{3}) = 3 \times 4\sqrt{3} = 12\sqrt{3}$.

why: When you subtract conjugates the rational parts cancel and the surd parts double - so only the surd survives.

- 3 Result: $\frac{12\sqrt{3}}{37}$.

why: Already rationalised, since the denominator is an integer.

Where it goes wrong

- $(2\sqrt{3})^2 = 4 \times 3 = 12$, not 6; using 6 gives denominator 43.
- Adding the numerators instead of subtracting, giving $\frac{42}{37}$ - which is offered.

Takeaway A difference of conjugate fractions leaves only the surd part; their product denominator is rational from the start.

Question 9 IND-22 · Indices and surds · spec MM1.2

Which one of the following is the largest?

Answer: A

Each option is $\sqrt{a} + \sqrt{b}$ with $a + b = 9$. Squaring gives $9 + 2\sqrt{ab}$, which is largest when the product ab is largest, i.e. when a and b are as close as possible. Option E has $a = b = 4.5$ (product 20.25), beating $4 \times 5 = 20$ for option C. Numerically: $E = \sqrt{18} \approx 4.243$, $C = 2 + \sqrt{5} \approx 4.236$.

Full method - how and why

A comparison question. Every option has the form $\sqrt{a} + \sqrt{b}$ with $a + b = 9$ - spotting that shared structure turns guesswork into a one-line argument.

- 1 Notice that in every option the two numbers under the roots add to 9 (including $2\sqrt{4.5} = \sqrt{4.5} + \sqrt{4.5}$).
why: When all the candidates share a constraint, compare them through that constraint rather than by estimating each one.
- 2 Square a general option: $(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab} = 9 + 2\sqrt{ab}$.
why: All the options are positive, so the largest option is the one with the largest square - and only the ab term varies.
- 3 With a fixed sum, ab is largest when a and b are closest together. The closest pair here is $a = b = 4.5$, giving $ab = 20.25$, beating $4 \times 5 = 20$.
why: This is the standard 'fixed perimeter, maximum area' fact; it decides the question without any decimals.
- 4 So $2\sqrt{4.5} = \sqrt{18} \approx 4.243$ is the largest (option C, $2 + \sqrt{5} \approx 4.236$, is a close second).
why: Worth a sanity check because the top two are so close.

Where it goes wrong

- Estimating roots to one decimal place - 4.24 versus 4.24 does not separate the top two options.
- Assuming $\sqrt{1} + \sqrt{8}$ is biggest because it contains the largest single number; the spread-out pair is actually the smallest.

Takeaway For a fixed sum $a + b$, the product ab - and hence $\sqrt{a} + \sqrt{b}$ - is maximised when a and b are as close as possible.

Question 10 X2-12 · Indices and surds · spec MM1.1

a and b are positive real numbers. Which of the following statements is/are true?

- I If $a^2 = b^2$ then $a = b$.
- II If $2^a = 2^b$ then $a = b$.
- III If $a^b = 1$ then $a = 1$.

Answer: H

I holds because $a, b > 0$ rules out $a = -b$. II holds because 2^x is one-to-one. III: for positive a and b , $a^b = 1$ gives $b \log a = 0$, so $\log a = 0$ and $a = 1$. All three are true.

Full method - how and why

Three cancellation claims, all made safe by the stated positivity. Check what each would need.

- 1 I: $a^2 = b^2$ gives $a = \pm b$ in general - but $a, b > 0$ rules out $a = -b$, so $a = b$. **True.**
why: The positivity hypothesis is exactly what repairs the usual counterexample.
- 2 II: 2^x is a strictly increasing function, so it takes each value once: $2^a = 2^b \Rightarrow a = b$. **True.**
why: One-to-one functions can be cancelled; equivalently take $\log_2$ of both sides.
- 3 III: $a^b = 1$ with $a, b > 0$. Take logs: $b \log a = 0$. Since $b > 0$, $\log a = 0$, so $a = 1$. **True.**
why: Taking logs turns the exponent into a factor, and a product is zero only if a factor is.
- 4 So all three.
why: Match to the list.

Where it goes wrong

- Rejecting I by recalling the counterexample 2 and -2 - forbidden here, since both are positive.
- Rejecting III by thinking of $a^0 = 1$ - that needs $b = 0$, which positivity forbids. (And $(-1)^2 = 1$ needs a negative a .)
- Overlooking how much work the phrase 'positive real numbers' is doing in all three.

Takeaway Read the domain restrictions first - positivity makes squaring and exponentiation reversible, when in general they are not.

Question 11 IND-12 · Indices and surds · spec MM1.1

For how many integer values of n with $1 \leq n \leq 100$ is $\sqrt{n} + \sqrt{n+1}$ rational?

Answer: D

For the sum to be rational, both $\sqrt{n}$ and $\sqrt{n+1}$ would need to be rational (if one is irrational, $\sqrt{n} + \sqrt{n+1}$ squared gives $2n+1 + 2\sqrt{n(n+1)}$ and $n(n+1)$ is never a perfect square for $n \geq 1$ since $n^2 < n(n+1) < (n+1)^2$). No two consecutive positive integers are both squares, so the answer is 0.

Full method - how and why

A 'for how many' question that is really about irrationality. The answer is 0, and seeing why needs one sharp observation about consecutive integers.

- 1 Suppose $\sqrt{n} + \sqrt{n+1} = r$ is rational and square it: $2n+1 + 2\sqrt{n(n+1)} = r^2$.
why: Squaring is the standard way to get at a sum of surds; the rational parts separate from the single remaining surd.
- 2 Rearrange: $\sqrt{n(n+1)} = \frac{r^2 - 2n - 1}{2}$, so $n(n+1)$ would have to be a perfect square.
why: A rational square root of an integer is an integer, so $n(n+1)$ must be a perfect square.
- 3 But $n^2 < n(n+1) < (n+1)^2$ for every $n \geq 1$, so $n(n+1)$ lies strictly between consecutive squares.
why: There is no integer square strictly between n^2 and $(n+1)^2$ - this is the contradiction.
- 4 Hence no n in $1 \leq n \leq 100$ works: the answer is 0.
why: The bound 100 is a red herring; the argument rules out every positive integer.

Where it goes wrong

- Testing $n = 3$ ($\sqrt{3} + 2$) and concluding 'sometimes it works' because one of the two roots is nice.
- Answering 9 or 10 by counting perfect squares up to 100 - that would answer a different question.

Takeaway $n(n+1)$ is never a perfect square for $n \geq 1$, because it is squeezed strictly between n^2 and $(n+1)^2$.